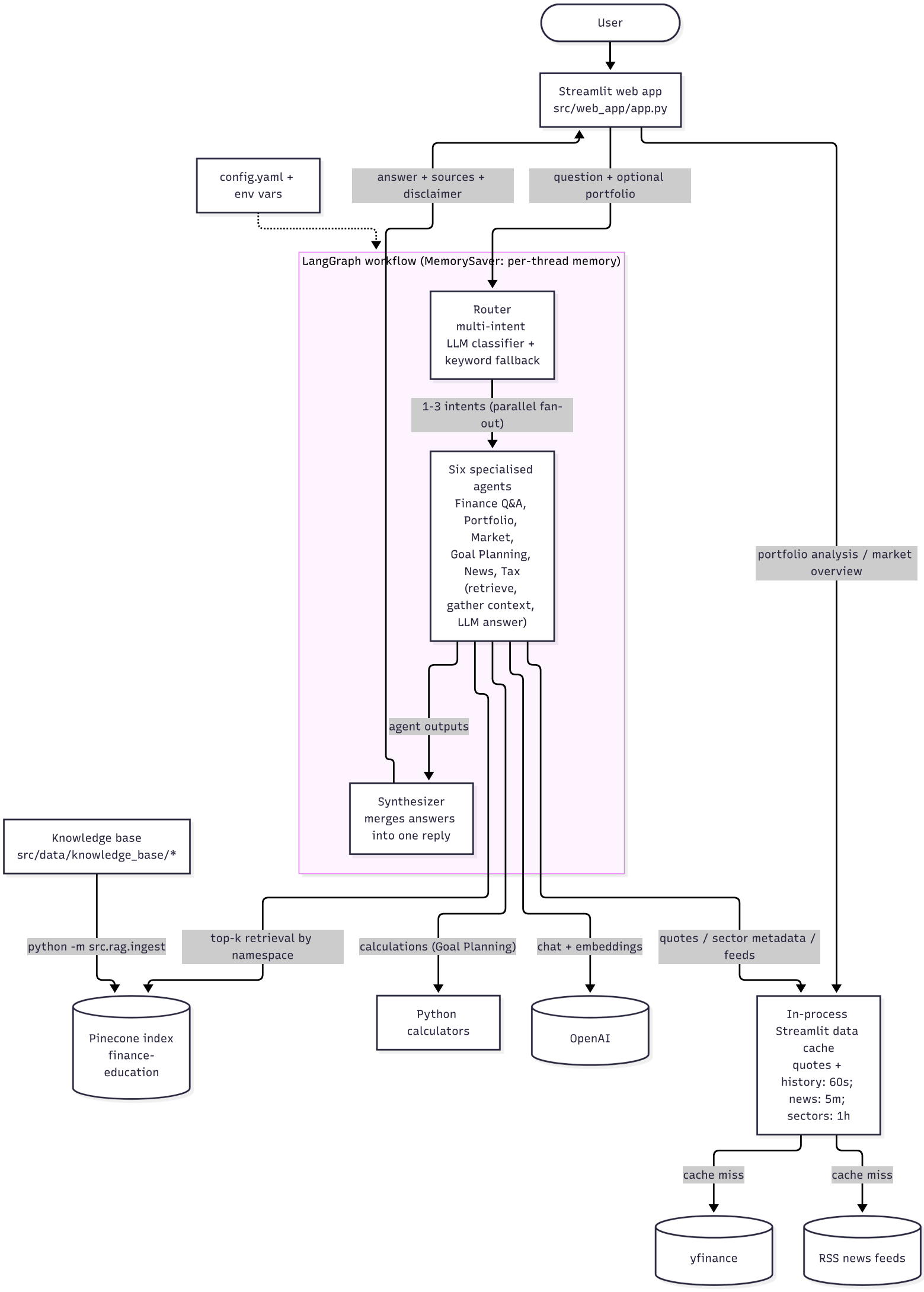
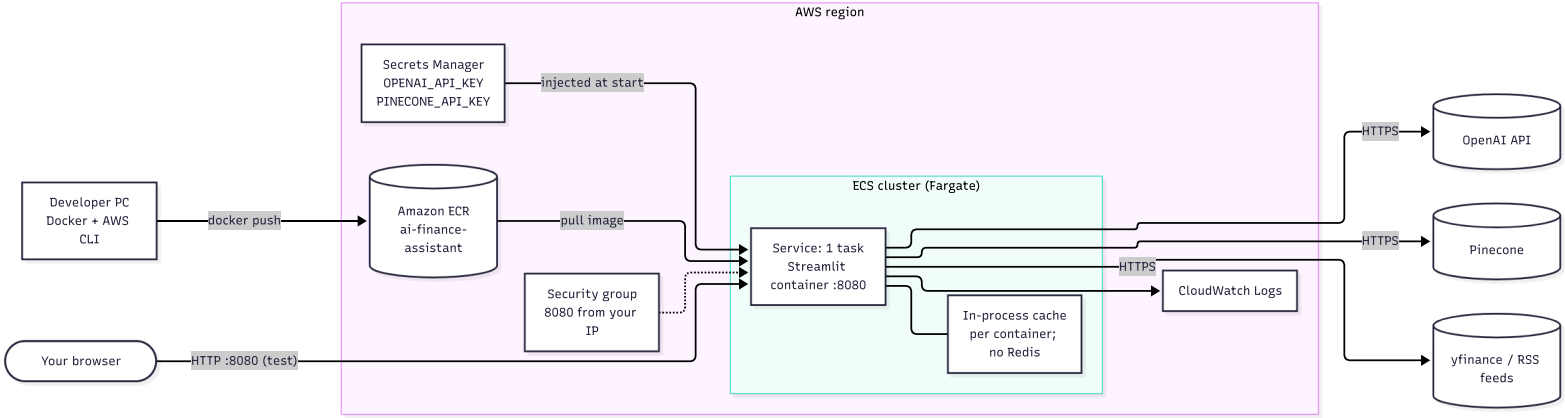

User
Streamlit web app
src/web_app/app.py
config.yaml +
env vars
answer + sources +
disclaimer
question + optional
portfolio
LangGraph workflow (MemorySaver: per-thread memory)
Router
multi-intent
LLM classifier +
keyword fallback
1-3 intents (parallel fan-
out)
Six specialised
agents
Finance Q&A,
Portfolio,
Market,
Goal Planning,
News, Tax
(retrieve,
gather context,
LLM answer)
agent outputs
Synthesizer
merges answers
into one reply
portfolio analysis / market
overview
Knowledge base
src/data/knowledge_base/*
python -m src.rag.ingest
top-k retrieval by
namespace
calculations (Goal Planning)
chat + embeddings
quotes / sector metadata /
feeds
Pinecone index
finance-
education
Python
calculators
OpenAI
In-process
Streamlit data
cache
quotes +
history: 60s;
news: 5m;
sectors: 1h
cache miss
cache miss
yfinance
RSS news feeds


AWS region
Secrets Manager
OPENAI_API_KEY
PINECONE_API_KEY
injected at start
Developer PC
Docker + AWS
CLI
docker push
Amazon ECR
ai-finance-
assistant
pull image
ECS cluster (Fargate)
Service: 1 task
Streamlit
container :8080
Security group
8080 from your
IP
In-process cache
per container;
no Redis
Your browser
HTTP :8080 (test)
HTTPS
OpenAI API
HTTPS
Pinecone
HTTPS
CloudWatch Logs
yfinance / RSS
feeds